Urinary Biomarkers Compendium

A cross-analyte synthesis of 31 single-biomarker literature reviews

Usense Healthcare — Data Science · 2026-04-17

Executive Summary

This compendium synthesises the thirty-one single-biomarker literature reviews produced in the `singleBiomarkers` knowledge-base project. Each review covers identity, physiology, clinical relevance, and the full spectrum of detection methodologies — from routine dipstick chemistry to state-of-the-art SERS and quantum-dot sensors. The present document distils that corpus into a cross-analyte view: classification, physicochemical range, detection-technology coverage, spectroscopic fingerprint atlas, and the sensitivity gap between clinical assays and emerging research methods.

Three headline findings emerge:

- **Urinalysis spans eight orders of magnitude in concentration**, from urea (~300 mM) down to PFAS (~1 nM). No single detection modality can cover this span alone — a hybrid sensing architecture is essential.
- **Spectrophotometry remains the most widely deployed clinical technique** (18 of 31 biomarkers), followed by electrochemistry (12) and HPLC/LC-MS (13). Fluorescence and SERS are the fastest-growing research frontiers, each covering 15+ biomarkers at the research stage.
- **Emerging methods deliver 10^2–10^4× sensitivity gains** over clinical gold standards. The largest gaps exist for urea (×17,606 with SERS), nitrites (×11,000), and glucose (×2,800) — suggesting where point-of-care miniaturisation could make the greatest clinical impact.

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1. Scope and Methodology

Each biomarker sheet in `outputs/` was generated to a strict template (Identity · Medical · Detection), and rendered to PDF individually. Structured data extracted from these sheets include chemical identity, typical urinary concentration, presence/absence in healthy urine, detection-method applicability, spectroscopic wavelengths and vibrational bands, and limits of detection (LOD). All cross-analyte figures in this document were built from that extracted dataset — see `generate_summary.py` in the repository root for the full data tables and plot code.

The 31 biomarkers were selected to cover the full clinical value of urinalysis at Usense:

- **Routine dipstick panel** — glucose, ketones, bilirubin, blood (RBC / haemoglobin), nitrites, leukocyte esterase, pH, specific gravity, protein (via phosphate/osmolality context).
- **Electrolytes & renal function** — sodium, chloride, phosphate, magnesium, urea, creatinine, uric acid, osmolality.
- **Metabolic auto-fluorophores** — NADH, FAD, riboflavin, tryptophan (intrinsic UV/blue fluorescence; probed by Usense's optical bench).
- **Porphyrin metabolites** — porphobilinogen, total urinary porphyrin (red fluorescence signature).
- **Stone-risk and organic-acid markers** — oxalate, citrate.
- **Trace metal** — copper (Wilson disease).
- **Environmental contaminants** — metolachlor, PFAS, chlorothalonil.

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2. Biomarker Landscape

2.1 Classification

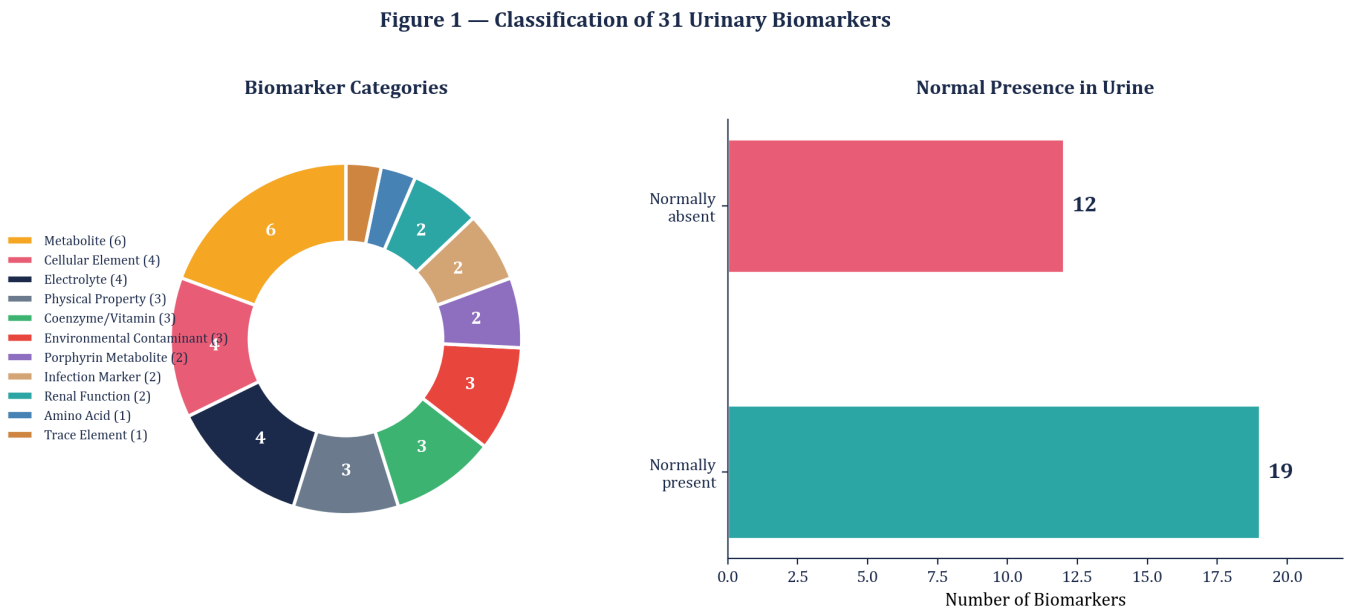

Figure 1: Figure 1 — Classification of 31 urinary biomarkers

Metabolites (6) form the largest category, followed by cellular elements and electrolytes (4 each). Crucially, **19 of the 31 analytes are normally present in urine** and must therefore be quantified; the other **12 are normally absent** (or present at trace levels), so their detection often triggers a clinical flag.

2.2 Molecular-weight spectrum

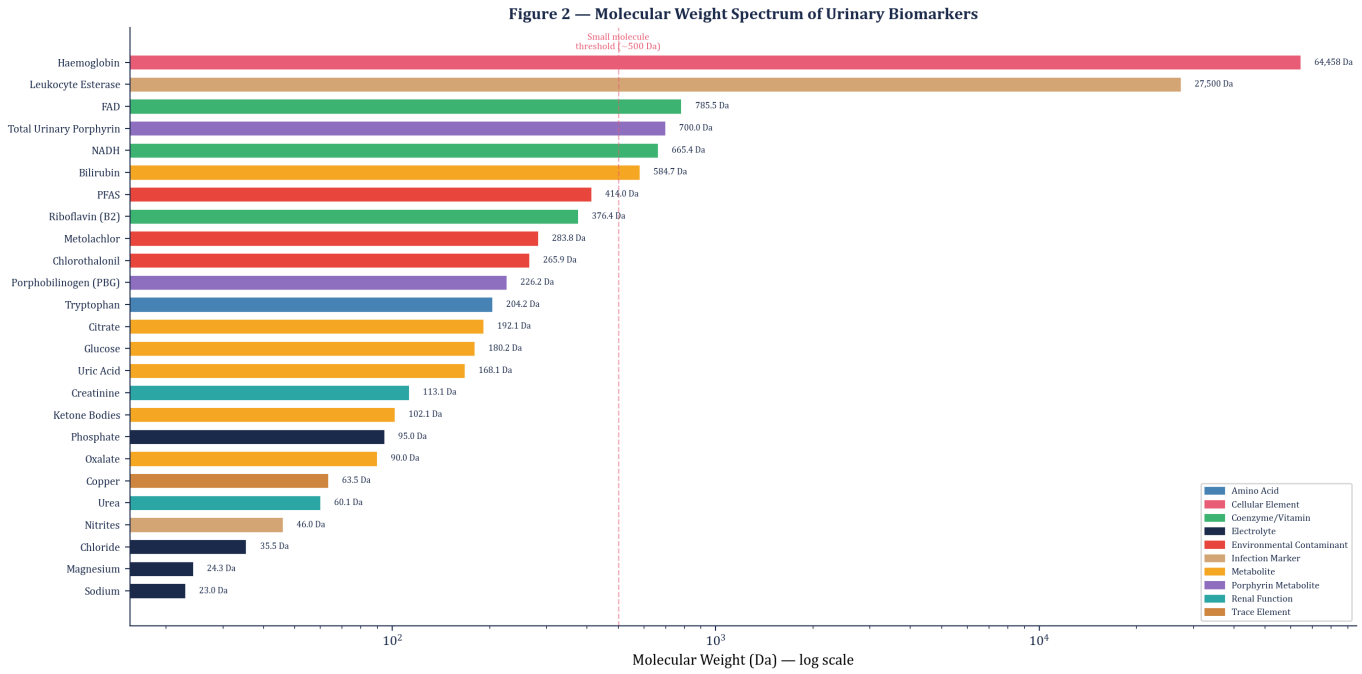

Figure 2: Figure 2 — Molecular weight spectrum of urinary biomarkers

The analyte molecular weights span more than three orders of magnitude, from sodium (23 Da) to haemoglobin (64.5 kDa). This has direct consequences for method selection: small ions (Na^+, Cl^-, Mg^{2+}) are ionically con-

ductive but optically invisible in the UV–visible; medium-sized conjugated molecules (bilirubin, FAD, riboflavin, porphyrins) are the natural targets for absorbance and fluorescence; large proteins (haemoglobin, leukocyte esterase) are best addressed by immunoassay or activity-based chemistry.

2.3 Urinary concentration ranges

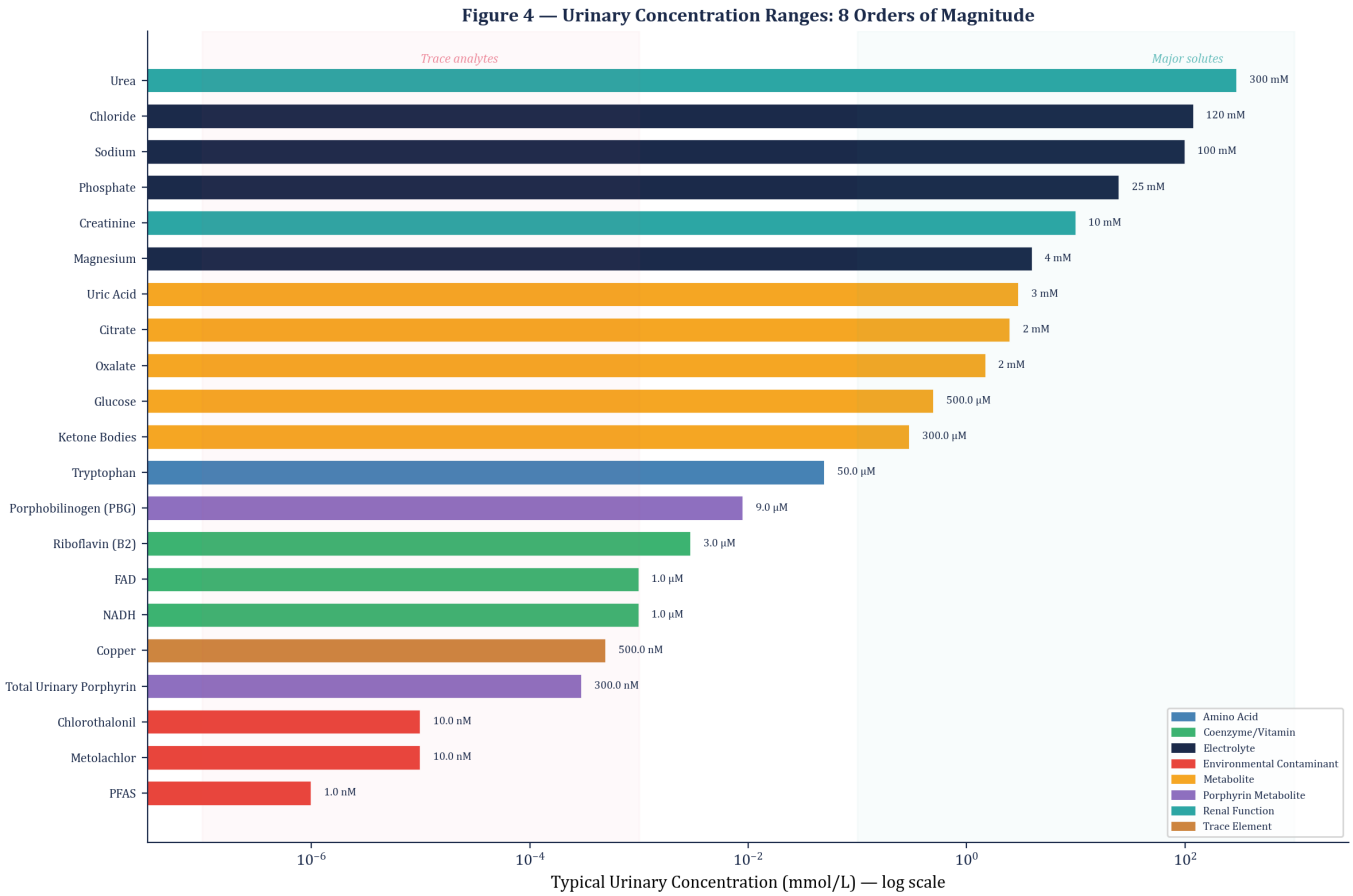

Figure 3: Figure 4 — Urinary concentration ranges span eight orders of magnitude

The concentration landscape is the single most important design constraint for a multianalyte sensor. Major solutes (urea, Na^+, Cl^-, phosphate, creatinine) sit in the 10–300 mM band, where tolerance to matrix effects outweighs raw LOD. Trace analytes (copper, porphyrins, environmental contaminants) require sub-nanomolar sensitivity that only HPLC-MS, SERS, or targeted immunosensing can reach. The dynamic range of **eight orders of magnitude** rules out any single linear transducer; tiered chemistry (enzymatic amplification, sample dilution, preconcentration) is mandatory.

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3. Detection Technology Landscape

3.1 Applicability matrix

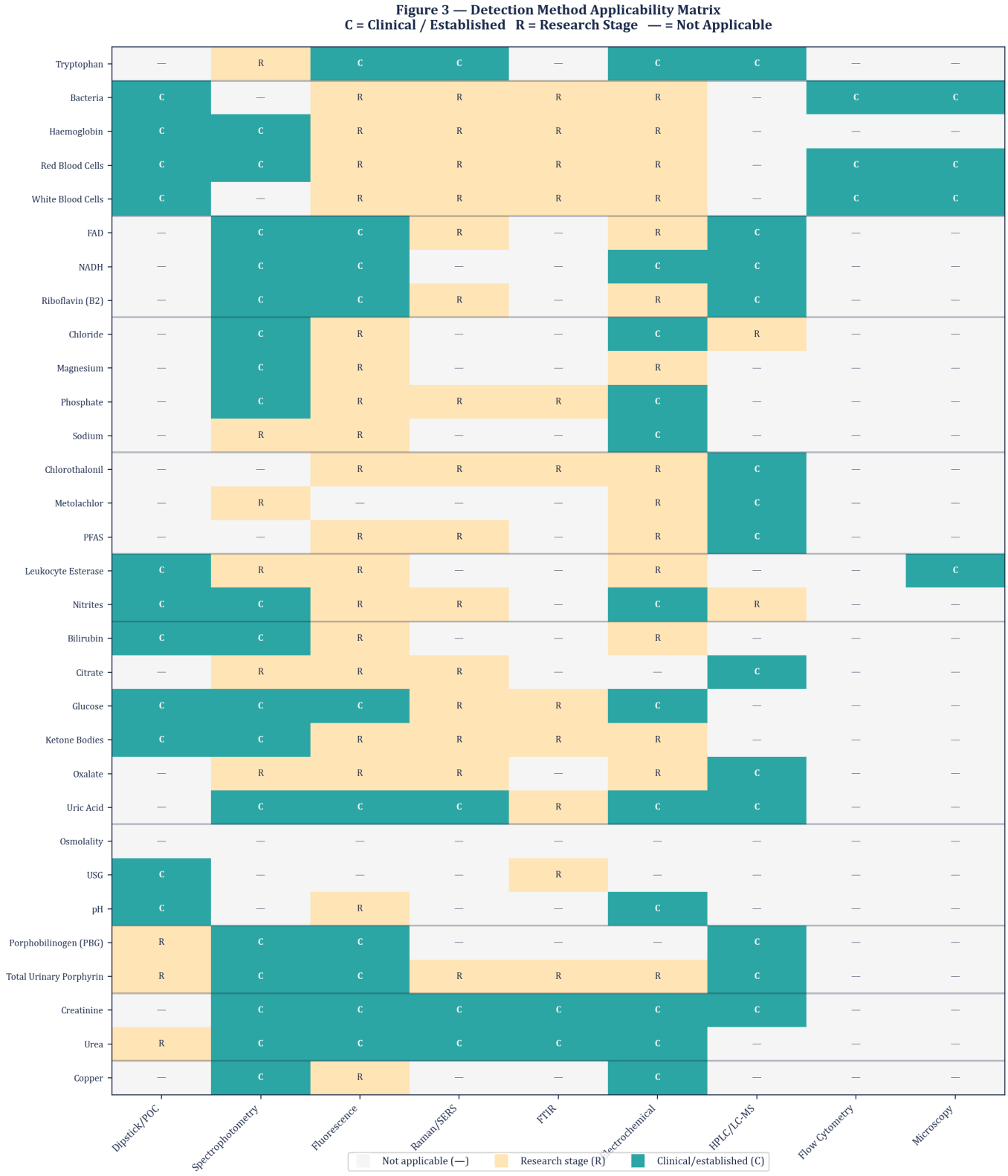

Figure 4: Figure 3 — Detection method applicability matrix

Each cell encodes whether a modality is **clinical / established (C)**, at **research stage (R)**, or **not applicable (—)** for a given biomarker. A few patterns stand out:

- **Spectrophotometry and electrochemistry** dominate the clinical column — they are the backbone of auto-

mated analysers and strip-based POC.

- **Fluorescence, Raman/SERS, and FTIR** are almost entirely in the research column. They promise reagent-free, multianalyte sensing but have yet to cross the clinical threshold for most targets.
- **HPLC/LC-MS** is clinically established for quantifying trace and regulatory-grade analytes (porphyrins, PFAS, pesticides, NADH/FAD in research contexts).
- **Flow cytometry and microscopy** are niche modalities restricted to cellular elements (RBCs, WBCs, bacteria, leukocyte esterase).

3.2 Coverage by technology

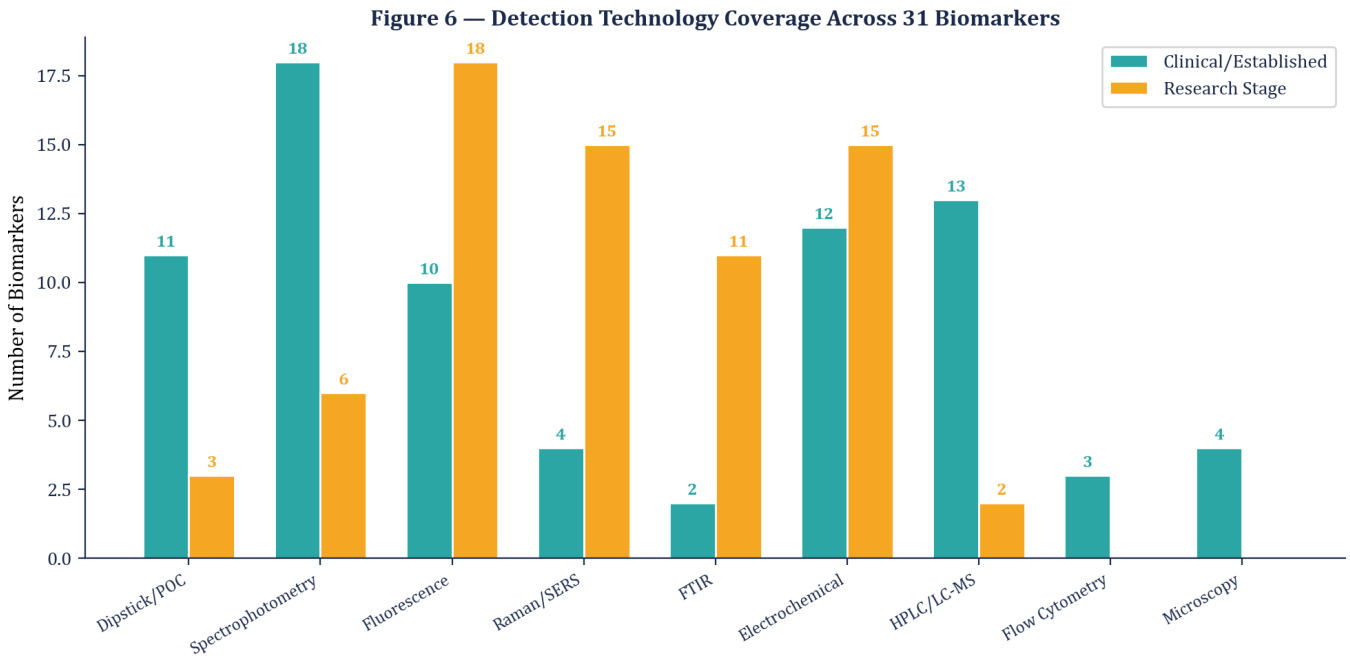

Figure 5: Figure 6 — Detection technology coverage across 31 biomarkers

Spectrophotometry covers the largest number of biomarkers at a clinical level (18), closely followed by HPLC/LC-MS (13) and electrochemistry (12). **Fluorescence and SERS** have the largest research-stage footprints (18 and 15 biomarkers respectively) — the most likely modalities to absorb new clinical targets in the next product cycle.

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4. Spectroscopic Fingerprint Atlas

4.1 Optical landscape (UV-Vis & fluorescence)

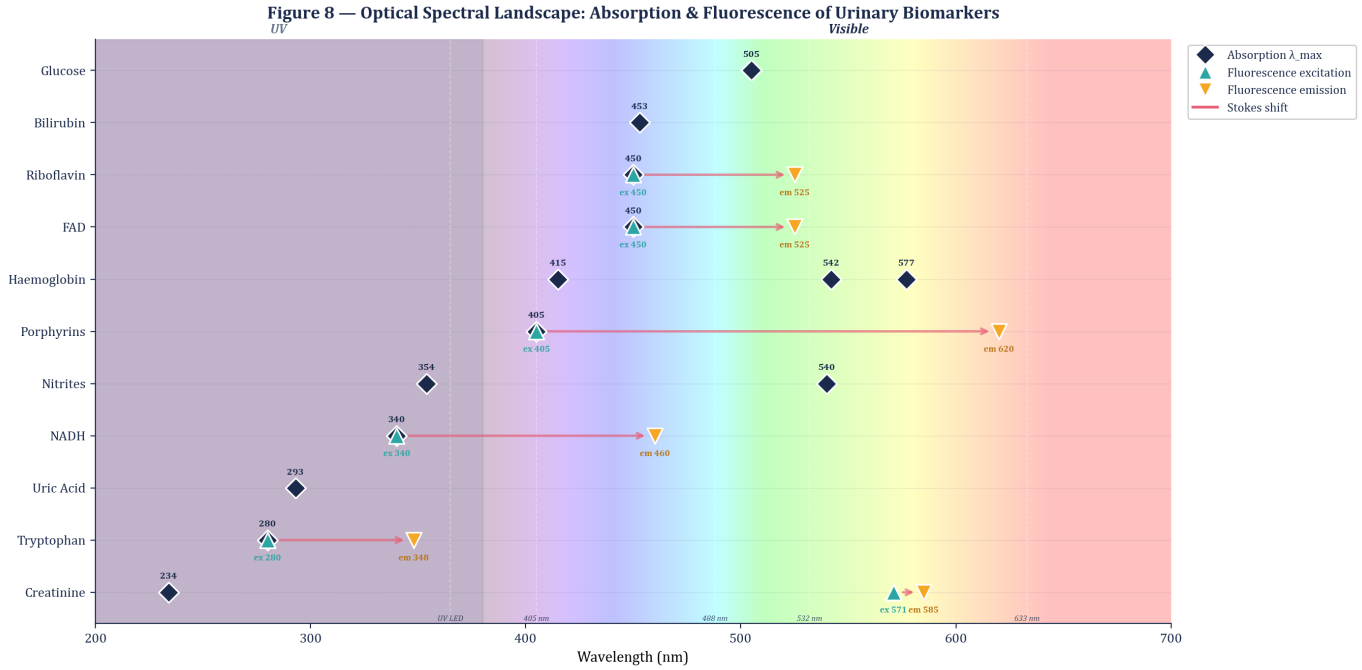

Figure 6: Figure 8 — Optical spectral landscape

This is the single most operationally-relevant chart in the compendium. Each biomarker's primary **absorption maximum** (dark diamond) is plotted against the visible spectrum; fluorescent species additionally show **excitation (teal up-triangle)** and **emission (amber down-triangle)** markers linked by a coral Stokes-shift arrow. Dashed vertical lines mark common laser / LED excitation sources.

Key observations:

- **Tryptophan** (280 nm abs, 348 nm em) is the sole strong native UV fluorophore and drives any deep-UV optical design.
- **NADH** (340 nm → 460 nm) is a highly stoked biomarker that pairs naturally with a 365 nm UV-LED.
- **Flavins (FAD, riboflavin)** share a tight 450 → 525 nm excitation-emission pair — the classic green-autofluorescence channel. A single 450 nm laser diode excites both with ≥525 nm emission separation.
- **Porphyrins** have a characteristic large Stokes shift (405 → 620 nm) that cleanly separates them from flavins on the same violet-excitation platform.
- **Haemoglobin**'s Soret band at 415 nm with Q-bands at 542/577 nm is uniquely specific and non-fluorescent — a strong absorbance discriminator.
- **Bilirubin** (453 nm) and **glucose-POD product** (505 nm) sit in the blue-green absorbance window, enabling white-LED spectrometry.

4.2 Vibrational fingerprint atlas (Raman & FTIR)

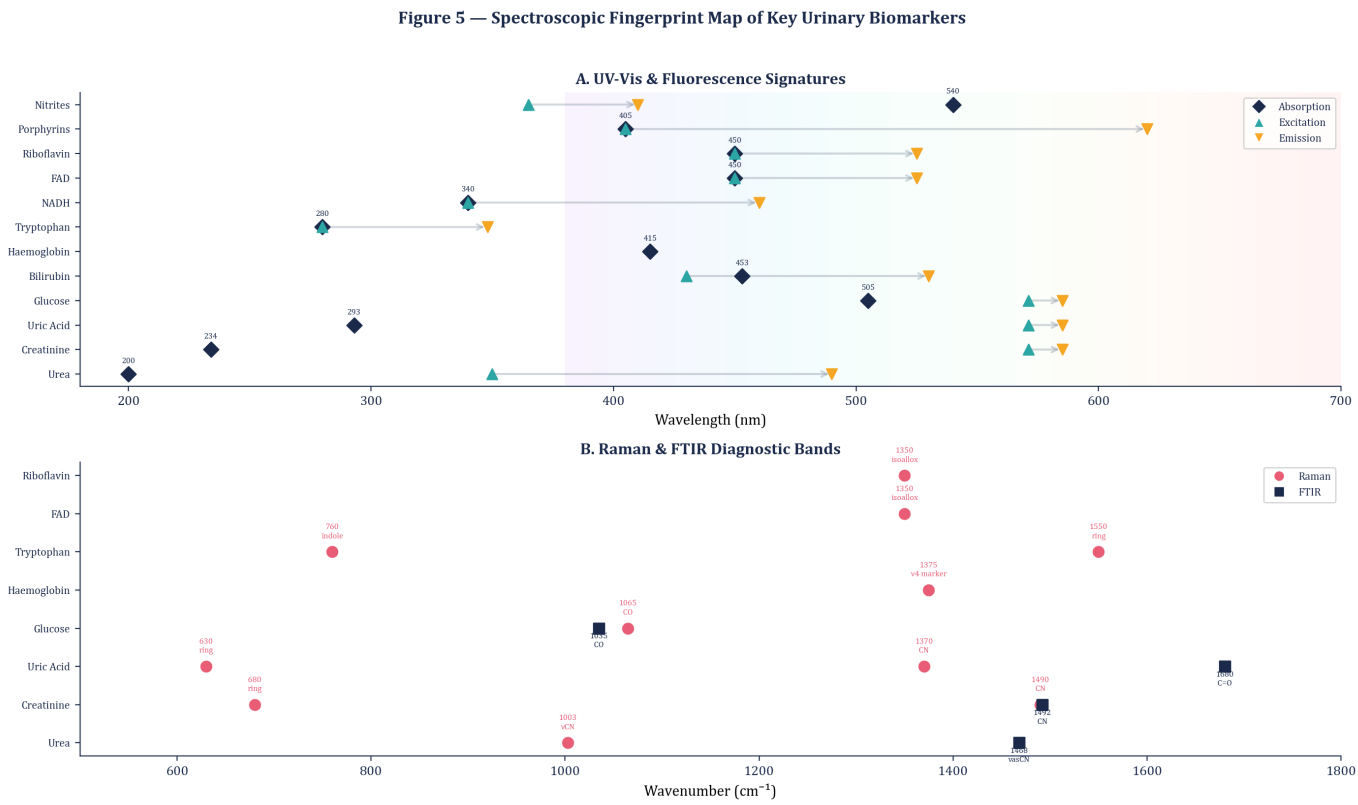

Figure 7: Figure 5 — Vibrational fingerprint atlas

Raman (Panel A) and FTIR (Panel B) probe orthogonal selection rules but share the **1000–1700 cm^{-1} biological fingerprint region** (shaded in teal). The most diagnostic bands are:

- **Urea** — ν(C–N) at 1003 cm^{-1} (Raman) and ν_as(C–N) at 1468 cm^{-1} (FTIR). The Raman band is uniquely narrow and far from other urinary species, making it the most accessible target for Raman-based urea quantification.
- **Creatinine** — 680 / 1490 cm^{-1} (Raman) and 1492 cm^{-1} (FTIR C=N) — complementary bands ideal for dual-modal detection.
- **Uric acid** — 630 cm^{-1} ring-breathing mode is uniquely low and isolated, and 1680 cm^{-1} C=O in FTIR provides secondary confirmation.
- **Nitrites, oxalate, citrate** — dominated by symmetric/asymmetric COO^- or NO_2 stretches that are strong in both Raman and FTIR, making ATR-FTIR a plausible multianalyte platform for organic-acid panels.
- **Flavins and porphyrins** — share isoalloxazine and Cβ-Cβ ring modes near 1350–1580 cm^{-1} that enable SERS discrimination at picomolar sensitivity.

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5. The Sensitivity Frontier

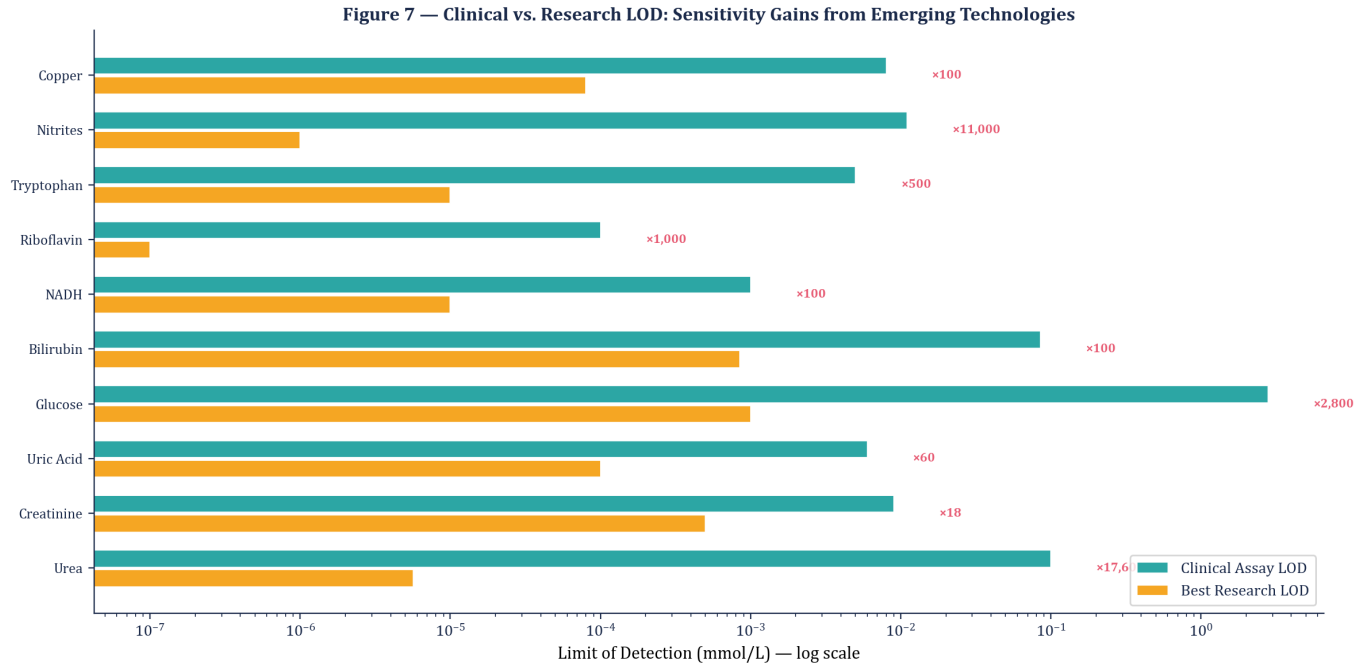

Figure 8: Figure 7 — Clinical vs. research LOD: sensitivity gains from emerging technologies

Figure 7 compares the best-of-class **clinical assay LOD** (teal) against the best-reported **research LOD** (amber) for ten representative biomarkers. The improvement factor (×, coral) reveals the sensitivity headroom available from next-generation chemistry.

- **Urea (×17,606)** — SERS on Fe_3O_4@C@Ag nanostructures reaches 5.7 nM vs. ~0.1 mM for the urease-GLDH gold standard. This headroom is larger than typical urinary urea concentration, implying that SERS could enable drop-based quantification without dilution.
- **Nitrites (×11,000)** — SERS and fluorescence assays reach picomolar ranges vs. micromolar for Griess. Critical for early UTI detection before bacterial overgrowth.
- **Glucose (×2,800)** — fluorescence and SERS LODs of ~1 μM compare with ~2.8 mM for GOD-POD strips. Real-world gain is limited by urinary glucose variance, but meaningful for diabetic micro-urinary-glucose tracking.
- **Riboflavin (×1,000)** — LIF-CE reaches 100 pM; the physiological urinary baseline (~3 μM) is well above, so the gain translates directly into sub-second integration times.
- **Creatinine (×18)** — the smallest gap: existing Jaffé and enzymatic assays are already excellent, and the research frontier is diminishing-returns territory.

Strategic read: The largest sensitivity gaps sit in biomarkers where **urinary concentration is close to the clinical assay LOD**. For those analytes (urea in dilute urine, glucose in early DM, nitrites in early UTI, riboflavin as a hydration / compliance marker), switching to a next-generation transducer unlocks clinically new information, not just a cleaner measurement.

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6. Cross-Modality Design Implications for Usense

Synthesising the above figures and the underlying 31 sheets yields the following design principles for a multianalyte urinary panel:

1. **A dual optical-electrochemical platform is the minimum viable architecture.** Optical channels (UV-Vis 280–700 nm + fluorescence) cover ~20 of the 31 biomarkers natively; electrochemistry covers the remaining ionic / redox-active species (Na^+, Cl^-, phosphate, pH, urea, glucose, uric acid).
2. **A 405 nm excitation source unlocks three orthogonal signals** — porphyrin emission at 620 nm, flavin

emission at 525 nm, and haemoglobin Soret absorption. A single violet LED therefore pays for three biomarkers.

3. **A 365 nm source addresses NADH (em 460 nm)** for metabolic state and tryptophan (em 348 nm in near-UV). Combining 365 + 405 + 488 nm excitation yields a complete autofluorescence panel.
4. **ATR-FTIR is the most promising reagent-free multianalyte method** (urea, creatinine, glucose, uric acid, ketones, nitrites, oxalate, citrate — all with $>10^{-3}$ M bands). SERS gains sensitivity but loses throughput and reproducibility; FTIR wins for volume routine analysis.
5. **SERS should be reserved for ultra-trace targets** — pesticides, PFAS, sub-micromolar metabolites, early-infection nitrites. A SERS add-on channel maximises return on substrate cost.
6. **Dipstick chemistry remains irreplaceable** for rapid POC — leukocyte esterase, bilirubin, ketones, and blood are most reliably flagged by enzymatic strip reactions. The roadmap should augment, not replace, the paper strip.

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7. Document Inventory

The full corpus of 31 single-biomarker sheets is available in `outputs/*.md` (source) and `papers/*.pdf` (rendered). Summary sheets are grouped below by category:

Category	Biomarkers
Metabolites	glucose, ketone, bilirubin, oxalate, citrate, uric acid
Cellular elements	bacteria, red blood cells, haemoglobin, white blood cells
Electrolytes	sodium, chloride, phosphate, magnesium
Physical properties	pH, osmolality, USG (specific gravity)
Coenzymes / vitamins	NADH, FAD, riboflavin
Environmental contaminants	metolachlor, PFAS, chlorothalonil
Porphyrin metabolites	porphobilinogen, total urinary porphyrin
Infection markers	nitrites, leukocyte esterase
Renal function	urea, creatinine
Amino acid	tryptophan
Trace element	copper

Each sheet covers, at minimum: canonical identity (PubChem CID, SMILES, MW, structural formula); endogenous/exogenous origin; primary/secondary biological roles; catabolism and renal handling; normal serum and urinary reference ranges; factors modulating levels; associated pathologies; clinical assays with LOD and interferences; the current gold standard; and a research-methods inventory spanning UV-Vis, fluorescence, Raman/SERS, FTIR, voltammetry, and microfluidic / paper / MIP / thermal platforms. Every quantitative claim carries a peer-reviewed citation.

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8. Key Take-Aways

- **Concentration is king.** The eight-order-of-magnitude urinary dynamic range forces any realistic panel to combine *amplifying* chemistries (enzymatic, SERS) with *high-capacity* linear transducers (spectrophotometry, electrochemistry).
- **Auto-fluorescence pays for itself.** NADH, FAD, riboflavin, porphyrins, and tryptophan form a self-consistent metabolic fluorescence panel from two UV-violet LEDs — no reagents, no consumables.
- **Raman-active fingerprint bands exist for every major organic solute.** A well-calibrated ATR-FTIR or SERS channel extracts urea, creatinine, glucose, uric acid, ketones, nitrites, oxalate, and citrate from a single 5-µL droplet.

- **The research-stage LOD frontier offers $\times 10^2$–$\times 10^4$ gains** — concentrated in trace-concentration analytes where sensitivity translates directly into new clinical utility (early infection, early dysglycaemia, environmental exposure).
- **Clinical dipstick chemistry is a floor, not a ceiling.** Every sheet identifies both the routine strip and a more sensitive research method; the roadmap is to preserve the former's speed while capturing the latter's sensitivity through integrated optical/electrochemical add-ons.

Methods and Data Sources

All figures in this compendium were generated by `generate_summary.py` from structured data extracted verbatim from the 31 single-biomarker reviews in `outputs/`. The source literature for each data point is cited at the end of the corresponding biomarker sheet. All single-biomarker PDFs rendered by the `md2pdf` pipeline are archived in `papers/`.

Colour palette, typography, and branding follow the Usense navy / amber / teal design language. Plots were rendered with matplotlib 3.x (serif face: Cambria / Times New Roman). Wavelength-to-RGB mapping in Figure 8 follows a piecewise approximation of the CIE 1931 standard observer.

End of compendium — 31 biomarkers, 8 figures, 1 view.